Layer2 Governance

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Q. What is L2?

A: Not sure anymore, but we used to believe

that L2 is a scaling solution to address the challenges of high fee and slow speed of L1 transactions by enabling off-(L1) chain transactions and making fewer L1 transactions that record only the final states as a result of many off-(L1) chain transactions

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“L2” might be a misleading marketing word

People might call anything “L2” when it satisfied most of the following conditions.

- Enable faster transactions with lower fee than on L1
- Implement the same virtual machine as L1, which enables the development of applications and protocols with the use of the same language as L1
- Have a mechanism to validate off-(L1) transactions at the time of exit and permanently store all off-(L1) transactions somewhere

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“L2” might be a misleading marketing word

“L2” tends to have many other characteristics beyond just a scaling solution.

- They allow users to make on-ramp and off-ramp transactions on “L2” without L1
- They enable engineers to develop applications and protocols on “L2” without L1
- They have their own governance (e.g., ‘decentralized’ organization, centralized entity)

Aren’t they L1s?
They call themselves “L2s” because it makes them sound as secure as their L1

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Again, Q: What is L2?

A: In this session, we define

L2 is a scaling solution that (1) **addresses the challenges of high fees and slow speeds of L1 transactions** by (2) **enabling 'off-L1 chain' transactions** and (3) **making fewer L1 transactions.** The L1 transactions are to (4) **record only the final states resulting from many 'off-L1 chain' transactions**, while ensuring that (5) **the integrity of all 'off-L1 chain' transactions remains intact until finality is achieved on L1**.

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The purpose of this project

- Analyzing various “L2s”, we identify their challenges for achieving condition (5) and demonstrate that most of them do not meet the definition of L2.
- We analyze risks involved in touchpoints of the system that lead to the challenges, and discuss technical measures and regulatory approaches to mitigate these risks.
- We analyze risks from regulatory perspectives:

  customer protection and **prevention of financial crimes and money laundering**

L2 definition in this project (repost of the previous page)

L2 is a scaling solution that (1) **addresses the challenges of high fees and slow speeds of L1 transactions** by (2) **enabling ‘off-L1 chain’ transactions** and (3) **making fewer L1 transactions.** The L1 transactions are to (4) **record only the final states resulting from many ‘off-L1 chain’ transactions**, while ensuring that (5) **the integrity of all ‘off-L1 chain’ transactions remains intact until finality is achieved on L1**.

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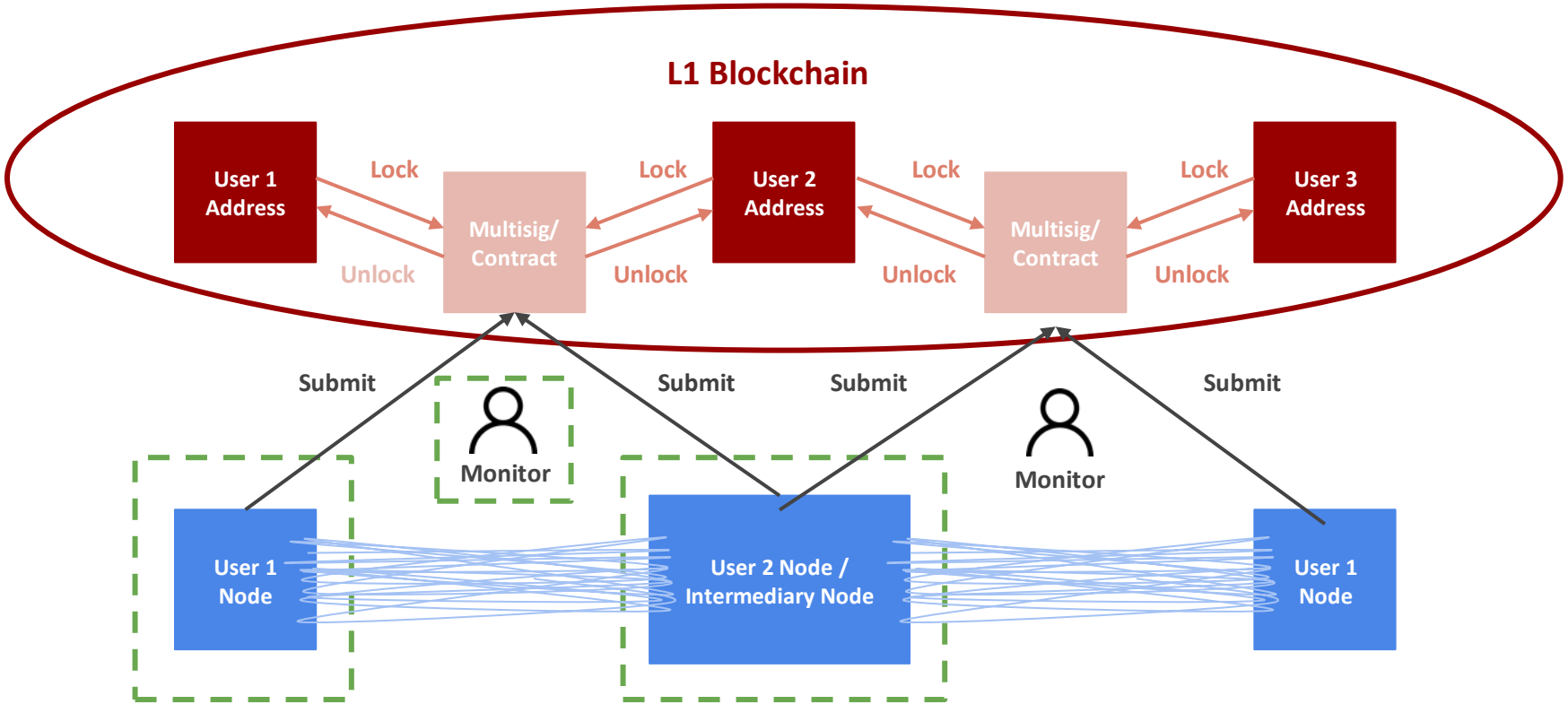
We analyzed the risks of Lighting Network



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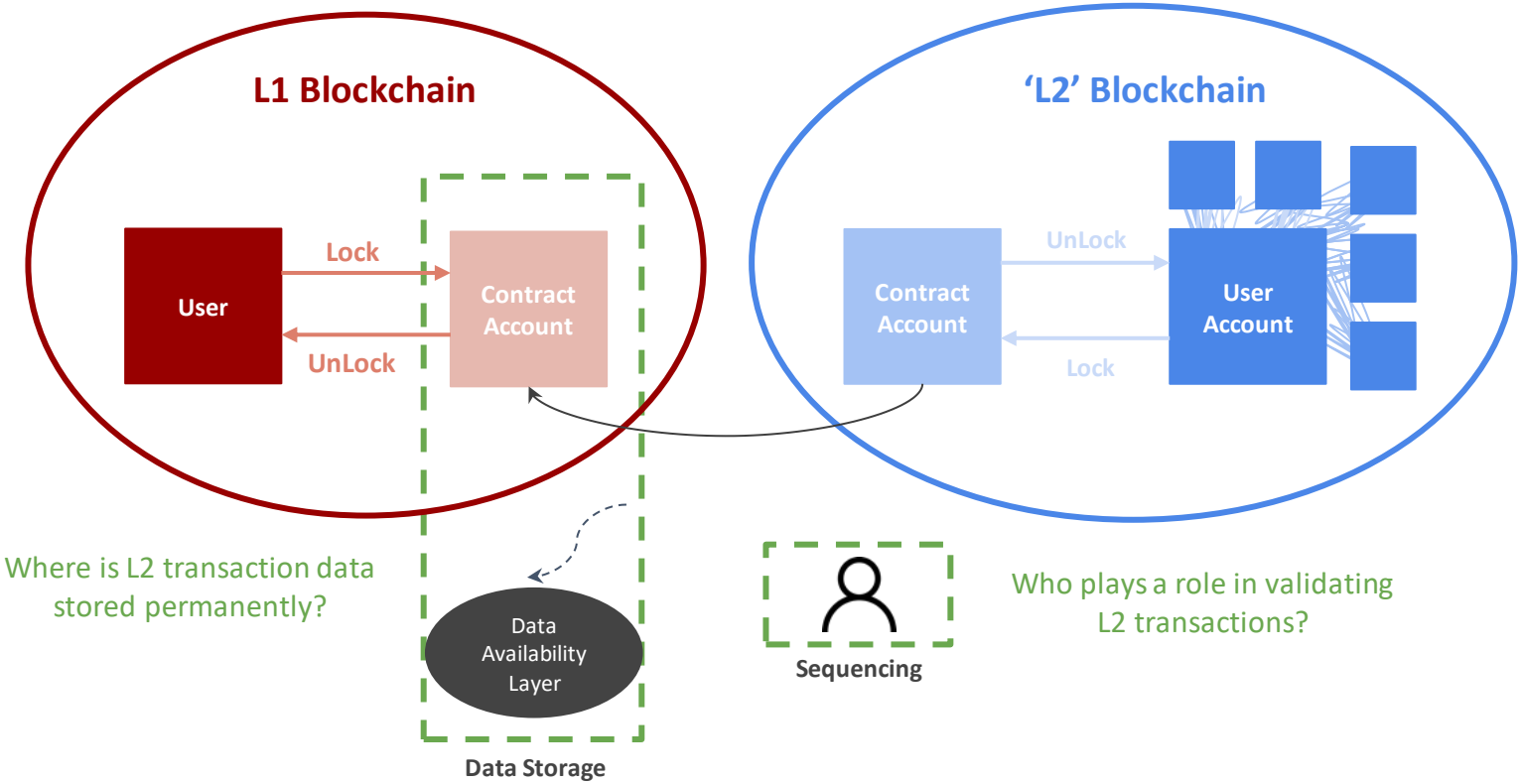
We analyze the risks of blockchain-based “L2s”



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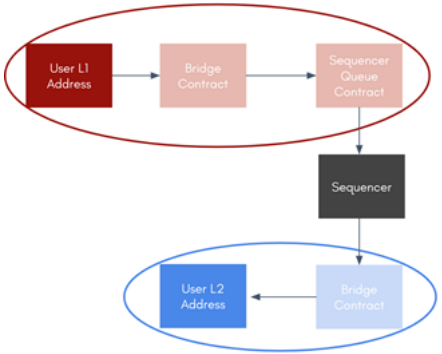
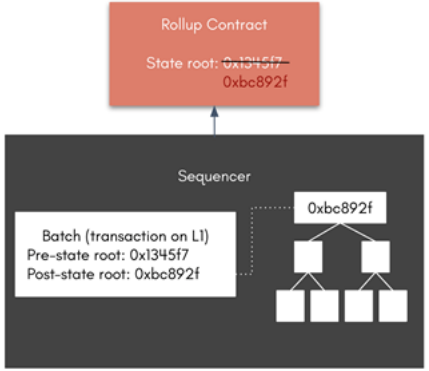
Regulatory touchpoint: Sequencing

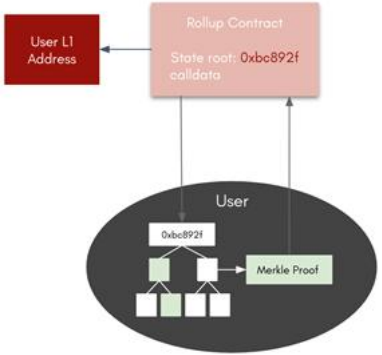
Ⅰ Entering phase



Ⅱ Using phase

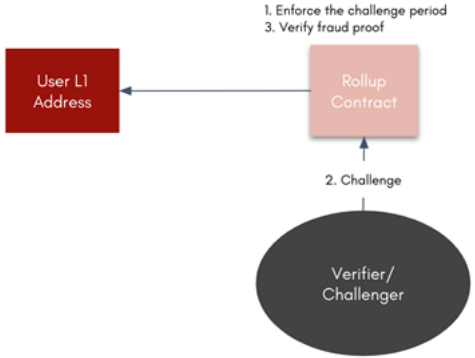


Ⅲ Exiting phase: Withdraw Request

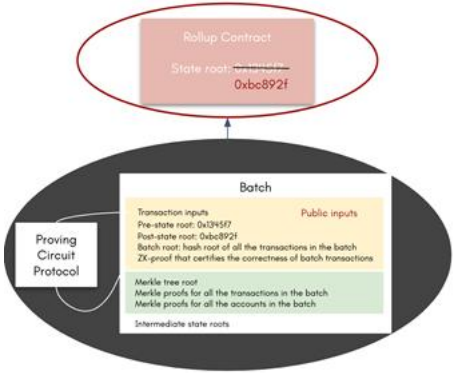


Ⅲ Exiting phase: Validation

Fraud proof



Validity proof (ZK proof)



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Risks involved in Sequencing

Validation methods	Fraud Proof	Validity Proof
Risks	• Smart contract failures and hacking • Users need to wait for the transaction finality on L1 until the challenge period ends • Users do not notice the improper work by sequencers and the challenge period ends, users lose their assets • If the number of sequencers are small, their centralization enable them to manipulate L2 transaction data • Proof of Commitment (PoC) > Collusion of Commitment (CoC)	• Smart contract failures and hacking • If the number of sequencers are small, their centralization enable them to intentionally not process certain transactions or delay them. • Due to the time required to generate ZK proofs, the immediacy of transactions can be compromised. When processing batches containing many transactions, the proof generation can take time, resulting in delays in transaction confirmation. • The role of generating ZK proofs (Prover) requires significant computational power, necessitating specialized hardware and optimized software for this purpose.

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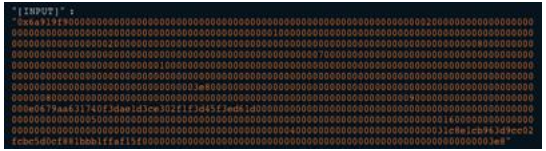
Regulatory touchpoint: Data Storage

Ⅰ Calldata

Calldata refers to the bytes sent as part of a transaction that are permanently recorded on- L1 chain. The amount of data stored in calldata directly impacts gas fees.

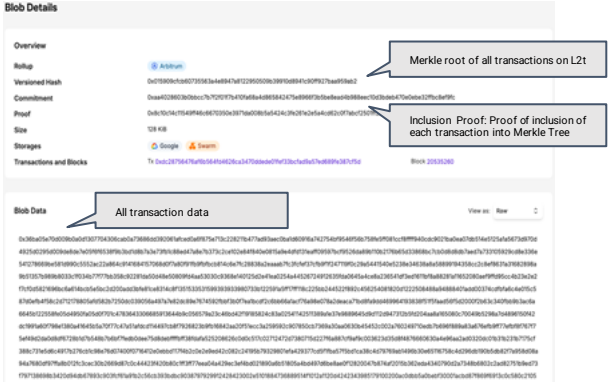
Cost	Security
△	◎



Ⅱ EIP-4844 (Blob)

A "Blob" is a new data format, designed to be handled differently from traditional transactions on Ethereum. Unlike standard transactions, Blobs are not stored directly on-chain; instead, they are transmitted and stored off-chain.

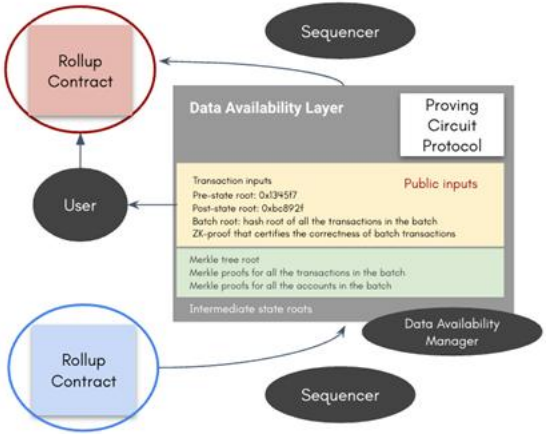
Cost	Security
○	○



Ⅲ Data Availability Layer

The off-L1 chain storage of L2 transaction data on data availability layer is ‘decentralized’ to prevent the Sequencer from compromising data availability through malicious actions. A group of trusted "Managers" is responsible for managing the storage and availability of L2 transaction data.

Cost	Security
◎	△



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Risks involved in Data Storage

Data storage	Calldata (on-L1 chain)	Blob/EIP-4844 (off-L1 chain)	Data Availability Layer (off-L1 chain)
Risks	• Using calldata costs gas fee associated with writing all L2 transactions on-chain	• Blob is a mechanism for temporary data storage, and once the verification process is complete, the data is deleted. Due to this temporary nature, there are security risks associated with data availability. • If the data is not properly maintained or if data availability is lost, there is a risk that users may not be able to retrieve the data. • Since blob is stored off-chain and only the commitment is saved on L1, there is a risk of **block withholding attacks**, where the data is intentionally concealed.	• Since all data is managed off-chain and the reliability of the data depends entirely on the Data Availability Committee (DAC), if the Data Availability Managers or Sequencers within the DAC are not sufficiently 'decentralized', it becomes easier for them to collude and carry out block withholding attacks. • If the Data Availability Managers are centralized, there are risks associated with the governance of DAC, as well as the potential for them to censor specific users or transactions. • In the event that the DAC fails to function properly, some users use an "escape hatch" mechanism that allows users to recover their assets. But it costs high gas fee and requires sophisticated infrastructure and software.

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Q. Is “governance” a magical tool?

- Let’s say that the definition of a ‘governance system’ is all about changing the rules of the game. Most typical way of changing the rules for L2s through a governance multi-sig or a issuance of governance tokens.
- ‘Governance’ can be sometimes seen as a magical tool to solve any problem; to fix bugs, to add features, to add incentives, etc. But we need to understand that governance comes with a major downside: It can be an attacker vector.

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